KAIZEN MATHS

Lesson starter

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

SA1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$0 \leq x < 10$	14	?
$10 \leq x < 15$	16	?
$15 \leq x < 20$	16	?
$20 \leq x < 25$	12	?
$25 \leq x < 35$	16	?
$35 \leq x < 40$	16	?

Show your working

KAIZEN MATHS

Lesson starter

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

SB1

Write the coordinates needed to draw the cumulative frequency curve.

Class interval	Frequency	Cumulative frequency
$30 \leq x < 40$	6	6
$40 \leq x < 45$	10	16
$45 \leq x < 55$	16	32
$55 \leq x < 65$	20	52
$65 \leq x < 75$	18	70
$75 \leq x < 85$	20	90

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KAIZEN MATHS

Lesson starter

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

SC1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$20 \leq x < 30$	8	?
$30 \leq x < 40$	6	?
$40 \leq x < 50$	10	?
$50 \leq x < 60$	20	?
$60 \leq x < 70$	12	?
$70 \leq x < 80$	20	?

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KAIZEN MATHS

Lesson starter

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

SD1

Write the coordinates needed to draw the cumulative frequency curve.

Class interval	Frequency	Cumulative frequency
$0 \leq x < 5$	16	16
$5 \leq x < 15$	20	36
$15 \leq x < 20$	10	46
$20 \leq x < 35$	18	64
$35 \leq x < 40$	18	82
$40 \leq x < 45$	8	90

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KAIZEN MATHS

Exit ticket

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

EA1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$10 \leq x < 25$	16	?
$25 \leq x < 35$	14	?
$35 \leq x < 45$	10	?
$45 \leq x < 50$	18	?
$50 \leq x < 60$	10	?
$60 \leq x < 75$	6	?

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KAIZEN MATHS

Exit ticket

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

EB1

Write the coordinates needed to draw the cumulative frequency curve.

Class interval	Frequency	Cumulative frequency
$10 \leq x < 20$	6	6
$20 \leq x < 35$	8	14
$35 \leq x < 45$	6	20
$45 \leq x < 50$	14	34
$50 \leq x < 60$	6	40
$60 \leq x < 70$	8	48

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KAIZEN MATHS

Exit ticket

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

EC1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$40 \leq x < 50$	20	?
$50 \leq x < 60$	6	?
$60 \leq x < 70$	10	?
$70 \leq x < 75$	18	?
$75 \leq x < 80$	20	?
$80 \leq x < 95$	18	?

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KAIZEN MATHS

Exit ticket

Cumulative Frequency Curves and Ogives

Tables & Coordinates

Name _____

EA1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$10 \leq x < 25$	16	?
$25 \leq x < 35$	14	?
$35 \leq x < 45$	10	?
$45 \leq x < 50$	18	?
$50 \leq x < 60$	10	?
$60 \leq x < 75$	6	?

Show your working

Teacher answers & working

Cumulative Frequency Curves and Ogives · Tables & Coordinates

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SA1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$0 \leq x < 10$	14	?
$10 \leq x < 15$	16	?
$15 \leq x < 20$	16	?
$20 \leq x < 25$	12	?
$25 \leq x < 35$	16	?
$35 \leq x < 40$	16	?

Answer:

0 ≤ x < 10 cumulative frequency 14; 10 ≤ x < 15 cumulative frequency 30; 15 ≤ x < 20 cumulative frequency 46; 20 ≤ x < 25 cumulative frequency 58; 25 ≤ x < 35 cumulative frequency 74; 35 ≤ x < 40 cumulative frequency 90

Working:

Cumulative frequency is a running total of the frequencies.

For 0 ≤ x < 10, add the frequency to the previous cumulative frequency.

$0 + 14 = 14$

For 10 ≤ x < 15, add the frequency to the previous cumulative frequency.

$14 + 16 = 30$

For 15 ≤ x < 20, add the frequency to the previous cumulative frequency.

$30 + 16 = 46$

For 20 ≤ x < 25, add the frequency to the previous cumulative frequency.

$46 + 12 = 58$

For 25 ≤ x < 35, add the frequency to the previous cumulative frequency.

$58 + 16 = 74$

For 35 ≤ x < 40, add the frequency to the previous cumulative frequency.

$74 + 16 = 90$

0 ≤ x < 10 cumulative frequency 14; 10 ≤ x < 15 cumulative frequency 30; 15 ≤ x < 20 cumulative frequency 46; 20 ≤ x < 25 cumulative frequency 58; 25 ≤ x < 35 cumulative frequency 74; 35 ≤ x < 40 cumulative frequency 90

Teacher answers & working

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SB1

Write the coordinates needed to draw the cumulative frequency curve.

Class interval	Frequency	Cumulative frequency
$30 \leq x < 40$	6	6
$40 \leq x < 45$	10	16
$45 \leq x < 55$	16	32
$55 \leq x < 65$	20	52
$65 \leq x < 75$	18	70
$75 \leq x < 85$	20	90

Answer:

(30, 0), (40, 6), (45, 16), (55, 32), (65, 52), (75, 70), (85, 90)

Working:

The first point is at the lower boundary of the first class with cumulative frequency 0.

first point = (30, 0)

Each other point uses the upper class boundary and the cumulative frequency.

upper boundary 40, cumulative frequency 6 gives (40, 6)

upper boundary 45, cumulative frequency 16 gives (45, 16)

upper boundary 55, cumulative frequency 32 gives (55, 32)

upper boundary 65, cumulative frequency 52 gives (65, 52)

upper boundary 75, cumulative frequency 70 gives (75, 70)

upper boundary 85, cumulative frequency 90 gives (85, 90)

(30, 0), (40, 6), (45, 16), (55, 32), (65, 52), (75, 70), (85, 90)

Teacher answers & working

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SC1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$20 \leq x < 30$	8	?
$30 \leq x < 40$	6	?
$40 \leq x < 50$	10	?
$50 \leq x < 60$	20	?
$60 \leq x < 70$	12	?
$70 \leq x < 80$	20	?

Answer:

20 ≤ x < 30 cumulative frequency 8; 30 ≤ x < 40 cumulative frequency 14; 40 ≤ x < 50 cumulative frequency 24; 50 ≤ x < 60 cumulative frequency 44; 60 ≤ x < 70 cumulative frequency 56; 70 ≤ x < 80 cumulative frequency 76

Working:

Cumulative frequency is a running total of the frequencies.

For 20 ≤ x < 30, add the frequency to the previous cumulative frequency.

$0 + 8 = 8$

For 30 ≤ x < 40, add the frequency to the previous cumulative frequency.

$8 + 6 = 14$

For 40 ≤ x < 50, add the frequency to the previous cumulative frequency.

$14 + 10 = 24$

For 50 ≤ x < 60, add the frequency to the previous cumulative frequency.

$24 + 20 = 44$

For 60 ≤ x < 70, add the frequency to the previous cumulative frequency.

$44 + 12 = 56$

For 70 ≤ x < 80, add the frequency to the previous cumulative frequency.

$56 + 20 = 76$

20 ≤ x < 30 cumulative frequency 8; 30 ≤ x < 40 cumulative frequency 14; 40 ≤ x < 50 cumulative frequency 24; 50 ≤ x < 60 cumulative frequency 44; 60 ≤ x < 70 cumulative frequency 56; 70 ≤ x < 80 cumulative frequency 76

Teacher answers & working

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SD1

Write the coordinates needed to draw the cumulative frequency curve.

Class interval	Frequency	Cumulative frequency
$0 \leq x < 5$	16	16
$5 \leq x < 15$	20	36
$15 \leq x < 20$	10	46
$20 \leq x < 35$	18	64
$35 \leq x < 40$	18	82
$40 \leq x < 45$	8	90

Answer:

(0, 0), (5, 16), (15, 36), (20, 46), (35, 64), (40, 82), (45, 90)

Working:

The first point is at the lower boundary of the first class with cumulative frequency 0.

first point = (0, 0)

Each other point uses the upper class boundary and the cumulative frequency.

upper boundary 5, cumulative frequency 16 gives (5, 16)

upper boundary 15, cumulative frequency 36 gives (15, 36)

upper boundary 20, cumulative frequency 46 gives (20, 46)

upper boundary 35, cumulative frequency 64 gives (35, 64)

upper boundary 40, cumulative frequency 82 gives (40, 82)

upper boundary 45, cumulative frequency 90 gives (45, 90)

(0, 0), (5, 16), (15, 36), (20, 46), (35, 64), (40, 82), (45, 90)

Teacher answers & working

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EA1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$10 \leq x < 25$	16	?
$25 \leq x < 35$	14	?
$35 \leq x < 45$	10	?
$45 \leq x < 50$	18	?
$50 \leq x < 60$	10	?
$60 \leq x < 75$	6	?

Answer:

10 ≤ x < 25 cumulative frequency 16; 25 ≤ x < 35 cumulative frequency 30; 35 ≤ x < 45 cumulative frequency 40; 45 ≤ x < 50 cumulative frequency 58; 50 ≤ x < 60 cumulative frequency 68; 60 ≤ x < 75 cumulative frequency 74

Working:

Cumulative frequency is a running total of the frequencies.

For 10 ≤ x < 25, add the frequency to the previous cumulative frequency.

$0 + 16 = 16$

For 25 ≤ x < 35, add the frequency to the previous cumulative frequency.

$16 + 14 = 30$

For 35 ≤ x < 45, add the frequency to the previous cumulative frequency.

$30 + 10 = 40$

For 45 ≤ x < 50, add the frequency to the previous cumulative frequency.

$40 + 18 = 58$

For 50 ≤ x < 60, add the frequency to the previous cumulative frequency.

$58 + 10 = 68$

For 60 ≤ x < 75, add the frequency to the previous cumulative frequency.

$68 + 6 = 74$

10 ≤ x < 25 cumulative frequency 16; 25 ≤ x < 35 cumulative frequency 30; 35 ≤ x < 45 cumulative frequency 40; 45 ≤ x < 50 cumulative frequency 58; 50 ≤ x < 60 cumulative frequency 68; 60 ≤ x < 75 cumulative frequency 74

Teacher answers & working

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EB1

Write the coordinates needed to draw the cumulative frequency curve.

Class interval	Frequency	Cumulative frequency
$10 \leq x < 20$	6	6
$20 \leq x < 35$	8	14
$35 \leq x < 45$	6	20
$45 \leq x < 50$	14	34
$50 \leq x < 60$	6	40
$60 \leq x < 70$	8	48

Answer:

(10, 0), (20, 6), (35, 14), (45, 20), (50, 34), (60, 40), (70, 48)

Working:

The first point is at the lower boundary of the first class with cumulative frequency 0.

first point = (10, 0)

Each other point uses the upper class boundary and the cumulative frequency.

upper boundary 20, cumulative frequency 6 gives (20, 6)

upper boundary 35, cumulative frequency 14 gives (35, 14)

upper boundary 45, cumulative frequency 20 gives (45, 20)

upper boundary 50, cumulative frequency 34 gives (50, 34)

upper boundary 60, cumulative frequency 40 gives (60, 40)

upper boundary 70, cumulative frequency 48 gives (70, 48)

(10, 0), (20, 6), (35, 14), (45, 20), (50, 34), (60, 40), (70, 48)

Teacher answers & working

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EC1

Complete the cumulative frequency column.

Class interval	Frequency	Cumulative frequency
$40 \leq x < 50$	20	?
$50 \leq x < 60$	6	?
$60 \leq x < 70$	10	?
$70 \leq x < 75$	18	?
$75 \leq x < 80$	20	?
$80 \leq x < 95$	18	?

Answer:

40 ≤ x < 50 cumulative frequency 20; 50 ≤ x < 60 cumulative frequency 26; 60 ≤ x < 70 cumulative frequency 36; 70 ≤ x < 75 cumulative frequency 54; 75 ≤ x < 80 cumulative frequency 74; 80 ≤ x < 95 cumulative frequency 92

Working:

Cumulative frequency is a running total of the frequencies.

For 40 ≤ x < 50, add the frequency to the previous cumulative frequency.

$0 + 20 = 20$

For 50 ≤ x < 60, add the frequency to the previous cumulative frequency.

$20 + 6 = 26$

For 60 ≤ x < 70, add the frequency to the previous cumulative frequency.

$26 + 10 = 36$

For 70 ≤ x < 75, add the frequency to the previous cumulative frequency.

$36 + 18 = 54$

For 75 ≤ x < 80, add the frequency to the previous cumulative frequency.

$54 + 20 = 74$

For 80 ≤ x < 95, add the frequency to the previous cumulative frequency.

$74 + 18 = 92$

40 ≤ x < 50 cumulative frequency 20; 50 ≤ x < 60 cumulative frequency 26; 60 ≤ x < 70 cumulative frequency 36; 70 ≤ x < 75 cumulative frequency 54; 75 ≤ x < 80 cumulative frequency 74; 80 ≤ x < 95 cumulative frequency 92